CS543 WEB SERVICES — ASSIGNMENT 2

# Smart Contact Manager (SCM)

Services, Contracts & Schema — Repository-Aligned Revision

| | |
|---|---|
| Team ID | 12 |
| Team members | Aman Bali — 20251651015; Bhupesh Kumar — 20251651032 |
| Project source | SERVICE_BASED repository uploaded for this assignment |
| Revision purpose | Correct service ports and SQL/MongoDB ownership to match the checked-in SCM repository |

This revision supersedes the earlier draft's proposed separate relational databases. It uses the checked-in MySQL migration as the relational-schema baseline and identifies MongoDB collections separately. Verify the live database and the `contact` versus JPA `contacts` table naming difference before applying migrations.

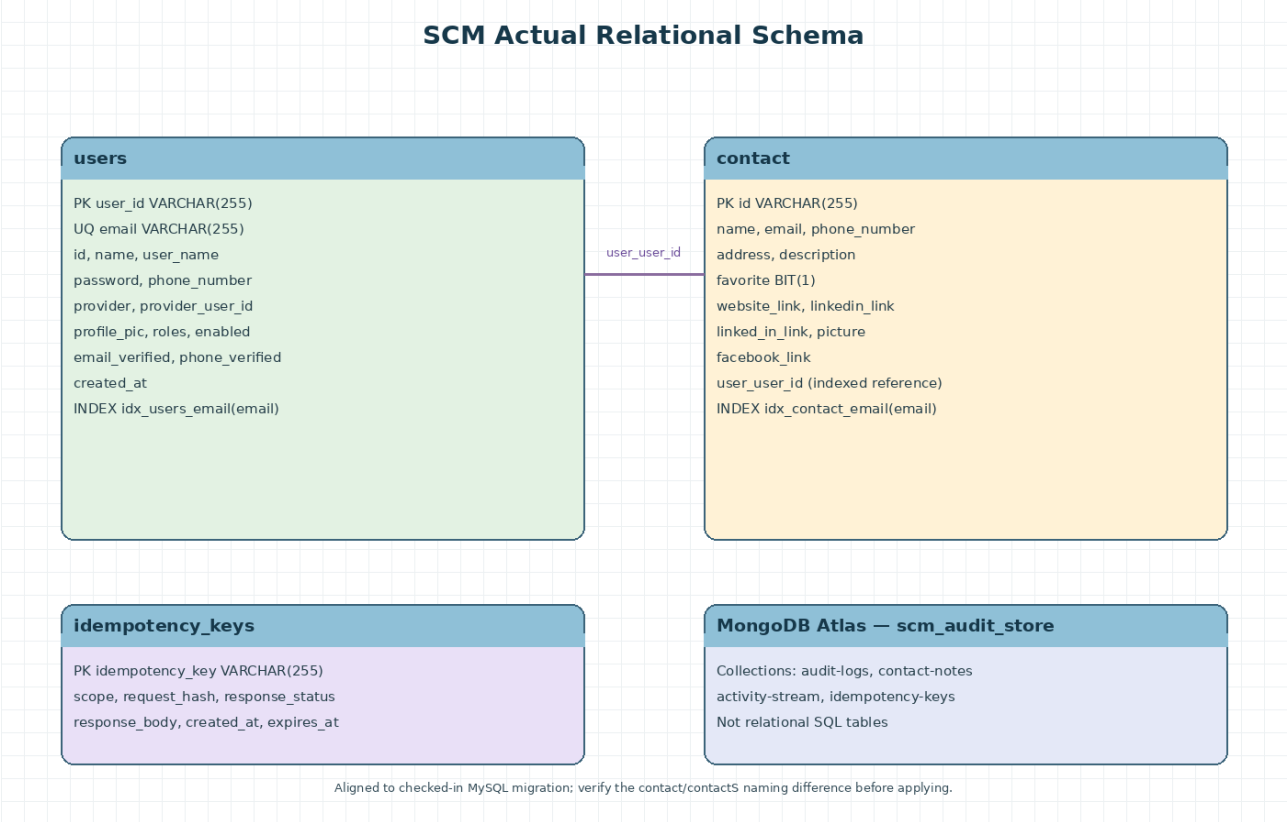
## 1. Service ownership

| Service | Port | Responsibility / boundary |
|---|---|---|
| API Gateway | 8080 | Express edge; routing, validation, OpenAPI/Swagger, GraphQL |
| Auth Service | 8081 | Identity, authentication, roles and OAuth |
| Contact Service | 8082 | Contact CRUD, search, favorite filtering and pagination |
| Media Service | 8083 | Contact photo/media upload and storage |
| Notification Service | 8084 | Asynchronous email dispatch |
| Audit Service | 8085 | MongoDB audit documents, contact notes and activity |
| Integration Service | 8086 | SOAP/WSDL bridge, timeout, retry and circuit breaker |

## 2. Persistence and schema ownership

The checked-in MySQL migration defines `users`, `contact`, and `idempotency_keys` in the same relational schema. MongoDB Atlas (`scm_audit_store`) stores `audit-logs`, `contact-notes`, `activity-stream`, and `idempotency-keys`. Media and notification responsibilities remain with their services; this document does not invent SQL tables and claim they are deployed.

# 3. Logical ER diagram




The diagram is aligned to the checked-in SQL migration. `user_user_id` is indexed as a reference; no cross-service foreign key is assumed. `schema.sql` includes the MySQL migration structures. MongoDB collections are listed separately because they are not relational SQL tables.

# 4. Contract boundaries

The public contract should describe business operations and stable response/error shapes, not expose table names or SQL. Gateway-to-service calls use HTTP contracts. Cross-service IDs are opaque references; do not add database foreign keys that cross service ownership boundaries.

# 5. Corrections made

• Replaced the earlier multi-database proposal with the repository's checked-in MySQL schema baseline.
• Corrected service ports to 8080–8086 as documented in the SCM README.
• Removed proposed media/notification SQL tables from the claimed deployed schema.
• Added a warning about `contact` in the migration versus `contacts` in the JPA entity mapping; verify the active schema before migration or deployment.